

Water Potability Prediction Using Machine Learning

Final Project Report

Machine Learning based Water Quality Classification System

1. Abstract

This project develops a machine learning based water quality classification system to predict whether a water sample is Potable (Safe to Drink) or Non-Potable (Not Safe to Drink). The project uses the Kaggle Water Potability Dataset containing 3276 samples and nine physicochemical water-quality features. Multiple classification algorithms were evaluated, including Random Forest, Support Vector Machine (SVM), Logistic Regression, Decision Tree, and K-Nearest Neighbors (KNN). A Hyperparameter-Tuned Random Forest was developed as the proposed model using GridSearchCV. An interactive Streamlit dashboard was also developed for dataset analysis, model evaluation, model comparison, and real-time prediction.

2. Problem Statement

Determining whether water is suitable for drinking is an important water quality classification problem. Manual analysis can require laboratory testing and expert interpretation. This project investigates whether machine learning can classify water samples using physicochemical measurements and provide an interactive prediction system.

3. Objectives

- Analyze the Water Potability Dataset.
- Perform exploratory data analysis and preprocessing.
- Handle missing values appropriately.
- Split the dataset into training, validation, and testing sets.
- Train and compare multiple machine learning classification models.
- Implement Random Forest and SVM as compulsory models.
- Develop a Hyperparameter-Tuned Random Forest as the proposed model.
- Evaluate models using Accuracy, Precision, Recall, F1-Score and ROC-AUC.
- Perform confusion matrix and 5-fold cross-validation analysis.
- Develop an interactive Streamlit dashboard for prediction.

4. Dataset

Property	Value
Dataset	Kaggle Water Potability Dataset
Total Samples	3276
Input Features	9
Total Columns	10

Target Column	Potability
Duplicate Rows	0
Classes	2

5. Input Features

The nine input features are: pH, Hardness, Solids, Chloramines, Sulfate, Conductivity, Organic Carbon, Trihalomethanes, and Turbidity.

Target: 0 = Non-Potable / Not Safe, 1 = Potable / Safe.

6. Data Preprocessing

The preprocessing workflow includes dataset loading, duplicate checking, missing-value analysis, missing-value imputation, train-validation-test splitting, and feature scaling. A preprocessing pipeline was saved using joblib so that the same transformations can be applied consistently to new prediction data.

7. Machine Learning Models

Model	Purpose
Random Forest	Compulsory baseline model
SVM	Compulsory classification model
Logistic Regression	Comparison model
Decision Tree	Comparison model
KNN	Comparison model
Tuned Random Forest	Proposed improved model

8. Proposed Model

The proposed model is a Hyperparameter-Tuned Random Forest. GridSearchCV was used to search for an effective combination of model hyperparameters.

Parameter	Selected Value
n_estimators	300
max_depth	10
max_features	sqrt
min_samples_leaf	1
min_samples_split	5

9. Validation Model Comparison

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Tuned Random Forest	0.6517	0.5714	0.4188	0.4834	0.6594
SVM	0.6273	0.5208	0.5236	0.5222	0.6570
Random Forest	0.6436	0.5588	0.3979	0.4648	0.6505
KNN	0.6314	0.5379	0.3717	0.4396	0.6286
Decision Tree	0.6090	0.4970	0.4346	0.4637	0.5773

Logistic Regression	0.5214	0.4141	0.5550	0.4743	0.5331
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The Hyperparameter-Tuned Random Forest achieved the highest validation accuracy and precision among the compared models and was selected as the proposed model.

10. Cross-Validation Results

Model	Mean ROC-AUC	Std. Deviation
SVM	0.7037	0.0140
Random Forest	0.6918	0.0174
Tuned Random Forest	0.6892	0.0213
KNN	0.6278	0.0113
Decision Tree	0.5563	0.0243
Logistic Regression	0.4855	0.0181

SVM achieved the highest mean cross-validation ROC-AUC. However, model selection was based on overall validation performance across multiple metrics rather than a single metric.

11. Final Test Results

Metric	Result
Accuracy	64.84%
Precision	55.76%
Recall	47.92%
F1-Score	51.54%
ROC-AUC	0.6758

12. Final Confusion Matrix

	Predicted Non-Potable	Predicted Potable
Actual Non-Potable	227	73
Actual Potable	100	92

The model correctly classified 227 non-potable samples and 92 potable samples. It incorrectly classified 73 non-potable samples as potable and 100 potable samples as non-potable.

13. Comparison with Base Paper

The base paper reports Random Forest and SVM as important classification models, with Random Forest performing better in its reported results. In this implementation, SVM can obtain stronger results on some evaluation measures. This difference is acceptable because machine learning results can vary with preprocessing, missing-value handling, data splitting, dataset distribution, hyperparameter settings, and evaluation methodology.

SVM achieved the highest 5-fold cross-validation ROC-AUC of 0.7037. However, SVM was not labelled as the proposed model. The proposed model is the Hyperparameter-Tuned Random Forest, selected based on overall validation performance.

14. Streamlit Dashboard

An interactive Streamlit dashboard was developed with five major sections: Dashboard, Dataset Summary, Final Model Evaluation, Model Comparison, and Water Potability Prediction.

Streamlit Dashboard

Navigation

Select Page

Dashboard

Dataset Summary

Model Performance

Model Comparison

Water Potability Prediction

Water Potability Prediction

Machine Learning based water quality classification system

Project Overview

This project predicts whether a water sample is potable or non-potable using physicochemical water-quality parameters.

Dataset Samples

3276

Input Features

9

Best Model

Tuned Random Forest

Target Classes

2

Project Information

Project: Water Potability Prediction

Dataset: Kaggle Water Potability Dataset

Primary Models: Random Forest and SVM

Best Model: Hyperparameter-Tuned Random Forest

Target: Potability

0 = Non-Potable / Not Safe

1 = Potable / Safe

Total Samples: 3276

Machine Learning Workflow

Dataset Collection → EDA → Missing Value Handling → Train/Validation/Test Split → Imputation → Feature Scaling → Model Training → Model Evaluation → Model Comparison → Best Model Selection → Final Prediction

Dataset Summary

Navigation

Select Page

Dashboard

Dataset Summary

Model Performance

Model Comparison

Water Potability Prediction

Water Potability Prediction

Machine Learning based water quality classification system

Dataset Summary

Dataset Preview

ph	Hardness	Solids	Chloramines	Sulfate	Conductivity	Organic_carbon	Trihalomethanes	Turbidity	Potability	
6	None	204.8965	26781.308	7.2003	368.5164	564.3887	10.3798	86.891	2.0621	0
7	3.7161	129.4232	19333.0579	6.6321	None	792.8554	13.18	56.3291	4.5077	0
7	8.6991	274.2181	19983.5427	9.7758	None	416.6867	16.0696	66.4091	3.6659	0
3	8.3368	214.3734	22018.4374	8.0593	256.8961	362.2655	10.4365	100.3417	4.6288	0
4	9.8927	181.1015	17616.3963	6.5466	518.1267	398.4109	11.5563	31.998	4.9151	0
5	5.5841	188.3133	26748.6077	7.5443	326.6764	280.4679	8.3597	54.9379	2.5597	0
6	10.7739	248.0177	28149.7193	7.5134	393.6634	783.6316	13.7897	94.4036	7.6173	0
7	6.4358	203.3615	13672.0918	4.963	353.3096	474.6076	12.3638	62.7983	4.4014	0
8	None	118.3686	34393.3639	7.8042	268.6469	389.3756	12.706	53.9389	3.195	0
9	11.1863	227.2383	25494.5085	9.0772	404.0416	563.8835	17.9278	71.8766	4.1706	0

Dataset Information

Rows

3276

Columns

10

Duplicate Rows

0

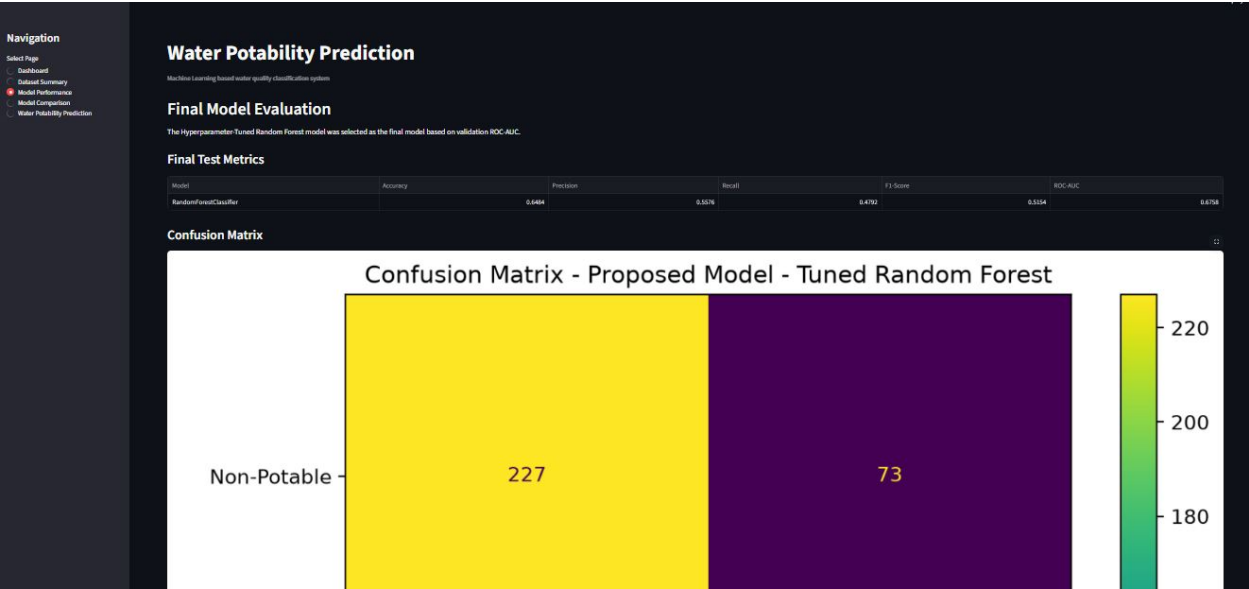
Missing Values

Feature	Missing Values
Sulfate	761
ph	491
Trihalomethanes	162

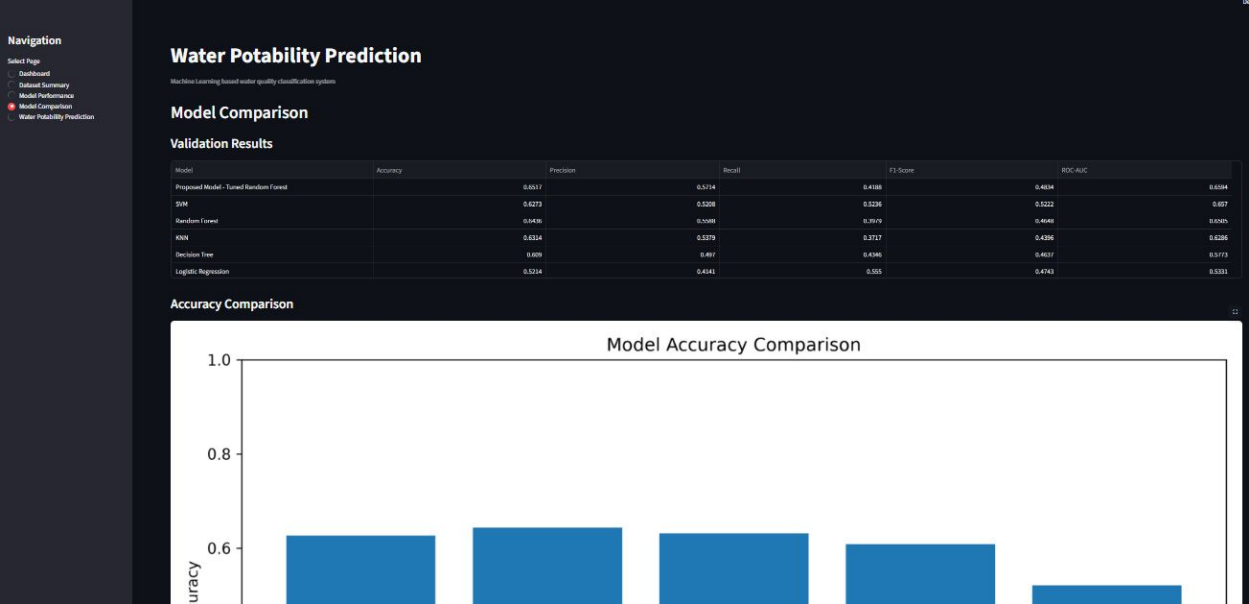
Target Distribution

Class	Count
Non-Potable	1998
Potable	1278

Final Model Evaluation



Model Comparison



Water Potability Prediction

Navigation

Select Page

- ☐ Dashboard
- ☐ Dataset Summary
- ☐ Model Performance
- ☐ Model Comparison
- ☒ Water Potability Prediction

Water Potability Prediction

Machine Learning based water quality classification system

Water Potability Prediction

Enter the physicochemical properties of the water sample to predict whether the water is potable.

pH	Chloramines	Organic Carbon
7.00	1.00	14.00
Hardness	Sulfate	TritationResiduals
196.00	103.00	95.00
Solids	Conductivity	Turbidity
2000.00	426.00	4.00

Predict Water Potability

Prediction Result

NOT SAFE TO DRINK

The model predicts that this water sample is non-potable.

Prediction Probability

Potable Probability

42.04%

Non-Potable Probability

57.96%

Water Potability Prediction Using Machine Learning | Random Forest and SVM based classification project

15. Prediction System

The prediction page accepts the nine physicochemical parameters: pH, Hardness, Solids, Chloramines, Sulfate, Conductivity, Organic Carbon, Trihalomethanes, and Turbidity. The trained proposed model and saved preprocessing pipeline are loaded from the models folder. The application returns a Potable or Non-Potable prediction together with potable and non-potable probabilities.

16. Project Structure

```
Water-Potability-ML/  
■■■■ app.py  
■■■■ README.md  
■■■■ requirements.txt  
■■■■ data/  
■■■■ models/  
■ ■■■■ best_model.joblib  
■ ■■■■ preprocessor.joblib  
■■■■ notebooks/  
■■■■ outputs/  
■■■■ screenshots/  
■■■■ src/  
■■■■ preprocessing.py  
■■■■ train_models.py  
■■■■ model_comparison.py  
■■■■ evaluate_models.py
```

17. Technologies Used

Python, Pandas, NumPy, Scikit-learn, Joblib, Matplotlib, and Streamlit were used to develop the project.

18. Conclusion

This project demonstrates an end-to-end machine learning workflow for water potability classification. Multiple classification models were implemented and compared using standard evaluation metrics. Random Forest and SVM were included as compulsory models, while a Hyperparameter-Tuned Random Forest was developed as the proposed model. The proposed model achieved 64.84% accuracy and 0.6758 ROC-AUC on the unseen test dataset. The Streamlit dashboard provides an interactive interface for dataset analysis, model evaluation, model comparison, and real-time water potability prediction.

Final Model: Hyperparameter-Tuned Random Forest

Final Test Accuracy: 64.84%

Final Test ROC-AUC: 0.6758